

Question 2 {3,4,7,8} 15

A)

i/j	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	T	F	F	T	F	F	F	F	F	F	F	F	F	F	F	F
4	T	F	F	T	T	F	F	T	F	F	F	F	F	F	F	F
7	T	F	F	T	T	F	F	T	F	F	T	T	F	F	T	F
8	T	F	F	T	T	F	F	T	T	F	T	T	T	F	T	T

B)

Start at A[3][15]=T

A[2][15]=F include S[3]=8 and then move to A[2][7]

A[1][7]=T exclude S[2]=7 and then move to A[1][3]

A[0][3]=F include S[1]=4 and then move to A[0][0]

j=3=S[0] include S[0] and move to A[0][0] Done

The included are 3,4,8 which 3+8+4 gives 15

T={3,4,8}

C)

Step 1

Start at A[3][15] = T

A[2][15] = F include s[3]=8, move to A[2][7]

A[1][7] = T exclude s[2]=7, move to A[1][3]

A[0][3] = F include s[1]=4, move to A[0][0]

j=3 = s[0]=3 include s[0]=3, move to A[0][0] DONE

Included: {3, 4, 8} 3 + 4 + 8 = 15

T = {3, 4, 8}

Step 2

Start at $A[3][15] = T$

$A[2][15] = F$ include $s[3]=8$, move to $A[2][7]$

$A[1][7] = T$ exclude $s[2]=7$ SPLIT!

$A[1][0] = T$ include $s[2]=7$, move to $A[1][0]$

$j=0$ DONE

Included: $\{7, 8\}$ $7 + 8 = 15$

$T = \{7, 8\}$

All Solutions = $\{ \{3, 4, 8\}, \{7, 8\} \}$

Question 3

i\w	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
a	0	0	0	0	0	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25	25
b	0	0	0	0	0	25	25	25	25	25	25	37	37	37	37	37	37	37	37	37	37
c	0	0	0	0	0	25	25	25	25	25	25	37	37	49	49	49	49	49	49	61	61
d	0	0	16	16	16	25	25	41	41	41	41	41	53	53	53	65	65	65	65	65	65
e	0	0	16	16	16	25	25	41	41	44	44	53	53	69	69	69	69	81	81	81	81

Maximum value = 81

Selected items = $\{a, b, d, e\}$

Total weight = $5 + 6 + 2 + 7 = 20$

Total value = $25 + 12 + 16 + 28 = 81$

Question 4

Capacity $W = 20$

Item	Value	Weight	Value/Weight
a	25	5	5.0
b	12	6	2.0
c	24	8	3.0
d	16	2	8.0
e	28	7	4.0

Sort by value/weight (descending): d, a, e, c, b

Selection:

Take d (2 weight, value 16)

Take a (5 weight, value 25)

Take e (7 weight, value 28)

Remaining capacity = $20 - (2+5+7) = 6$

Take $6/8$ of c $\rightarrow$ value = $24 * (6/8) = 18$

Final Answer:

Total Value = $16 + 25 + 28 + 18 = 87$